Carryover-sensitivity study: development-run review

96-cell grid, five analysis methods, 150 replicates per cell

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Contents

1 Overview

This document presents pipeline-verification results for the carryover-sensitivity manuscript from the development simulation run completed 2026-05-20. The run uses a 96-cell reduced Tier 1 grid (150 replicates per cell, MCSE ≈ 0.041 at $p = 0.5$) and a 31-cell Tier 2 sensitivity grid. All figures and tables are computed inline from the development RDS files; nothing is loaded from pre-rendered figures or the shared `analysis/data/` summaries.

Five analysis methods. Beyond the three analysis-model specifications of the parent study, E1 (binary D_b), E2 (exposure-weighted D_{bc}), and E3 (lagged-treatment), this dev run carries two robust-inference analyses, both applied to the E2 (D_{bc}) predictor. `lme+CR2` is the conditional `lme` fit with its model-based standard error replaced by a CR2 bias-reduced cluster-robust sandwich. `GEE+MD` is the marginal generalized-estimating-equations fit with a Mancl-DeRouen bias-corrected sandwich. The two robust analyses are produced by the `--robust` flag of the simulation drivers, in anticipation of carrying them into the production pipeline; their rationale and validation are documented in manuscript 10 (`analysis/report/10-interaction-test-calibration/`).

The purpose of this document is to verify that:

Table 1: Development-run metadata read from the output RDS files.

Item	Value
Tier 1 cells	96
Tier 1 reps/cell	150
Tier 2 cells	31
Tier 2 reps/cell	150
Analysis methods per cell	5
Total Tier 1 rows	72000
Total Tier 2 rows	23250
Run date	2026-05-21

a) the E2 > E1 ≈ E3 power ranking is visible at 150 reps;
b) the three model-based specifications are conservative under the null, while the robust analyses move the Type I error toward the nominal 0.05;
c) lme+CR2 recovers the power lost to conservatism, while GEE+MD carries a modest residual efficiency cost as a marginal-model estimator;
d) the Tier 2 sensitivity-block trends are plausible.

Precise point estimates should be drawn from the production run (500 reps, 540 cells) rather than from this document.

Dev grid coverage. The Tier 1 dev grid retains both DGP architectures and all three DGP decay forms (the primary axes), trimming secondary axes to two representative values each. `t1half` is reduced to $\{0.0, 1.0\}$ weeks; design to $\{\text{CO}, \text{Hybrid}\}$; N to $\{35, 70\}$; c_{bm} to $\{0.0, 0.45\}$; Weibull uses only $k = 0.7$. The OL+BDC design and $k = 1.5$ Weibull shape are absent from the dev grid; any figure filtered to those levels will appear empty.

2 Tier 1 results

2.1 Headline performance at the reference cell

Table 2 shows the Morris-style performance measures at the reference cell (the covariance-moderation architecture, exponential DGP, Hybrid design, $N = 70$, $c_{bm} = 0.45$), for all five analysis methods at both carryover half-lives present in the dev grid.

2.2 Power by analysis method

2.3 Decay-form by analysis-method heatmap

2.4 Type I error control

The three model-based specifications E1, E2, and E3 sit visibly below the nominal 0.05 line in Figure 3, the conservatism diagnosed in manuscript 10 (`analysis/report/10-interaction-test-calibration/`); lme+CR2 and GEE+MD sit on or near it, confirming that the robust analyses recalibrate the interaction test. Figure 4 shows the same null cells split by DGP architecture rather than by design. Because the two architectures reduce to a bit-identical data-generating process when $c_{bm} = 0$, the Mean and Covariance panels are two samples from one distribution and should coincide up to Monte Carlo noise; that they do is a consistency check on the simulation.

2.5 Summary table: all cells at $N = 70$, $c_{bm} = 0.45$, Hybrid

Table 2: Reference-cell performance (the covariance-moderation architecture, exponential DGP, Hybrid, $N = 70$, $c_{bm} = 0.45$, $n_{sim} = 150$). The lme+CR2 row shares the E2 interaction estimate (its bias and empirical SE track E2) and differs only in the standard error; GEE+MD is a separate marginal fit with its own estimate, modestly less efficient than the conditional lme. MCSEs are about twice those of the 500-rep production run.

t1/2	spec	Power (MCSE)	Bias (MCSE)	Emp SE	Coverage	Non-conv
0	E1	0.967 (0.015)	+0.043 (0.071)	0.870	0.953	0.000
0	E2	0.967 (0.015)	+0.043 (0.071)	0.870	0.953	0.000
0	E3	0.967 (0.015)	+0.044 (0.071)	0.866	0.953	0.000
0	lme+CR2	0.987 (0.009)	+0.043 (0.071)	0.873	0.906	0.007
0	GEE+MD	0.960 (0.016)	+0.095 (0.074)	0.908	0.913	0.000
1	E1	0.727 (0.036)	+1.231 (0.065)	0.792	0.767	0.000
1	E2	0.873 (0.027)	+0.098 (0.081)	0.997	1.000	0.000
1	E3	0.720 (0.037)	+1.249 (0.066)	0.804	0.767	0.000
1	lme+CR2	0.893 (0.025)	+0.095 (0.082)	0.999	0.946	0.007
1	GEE+MD	0.847 (0.029)	+0.068 (0.086)	1.058	0.960	0.000

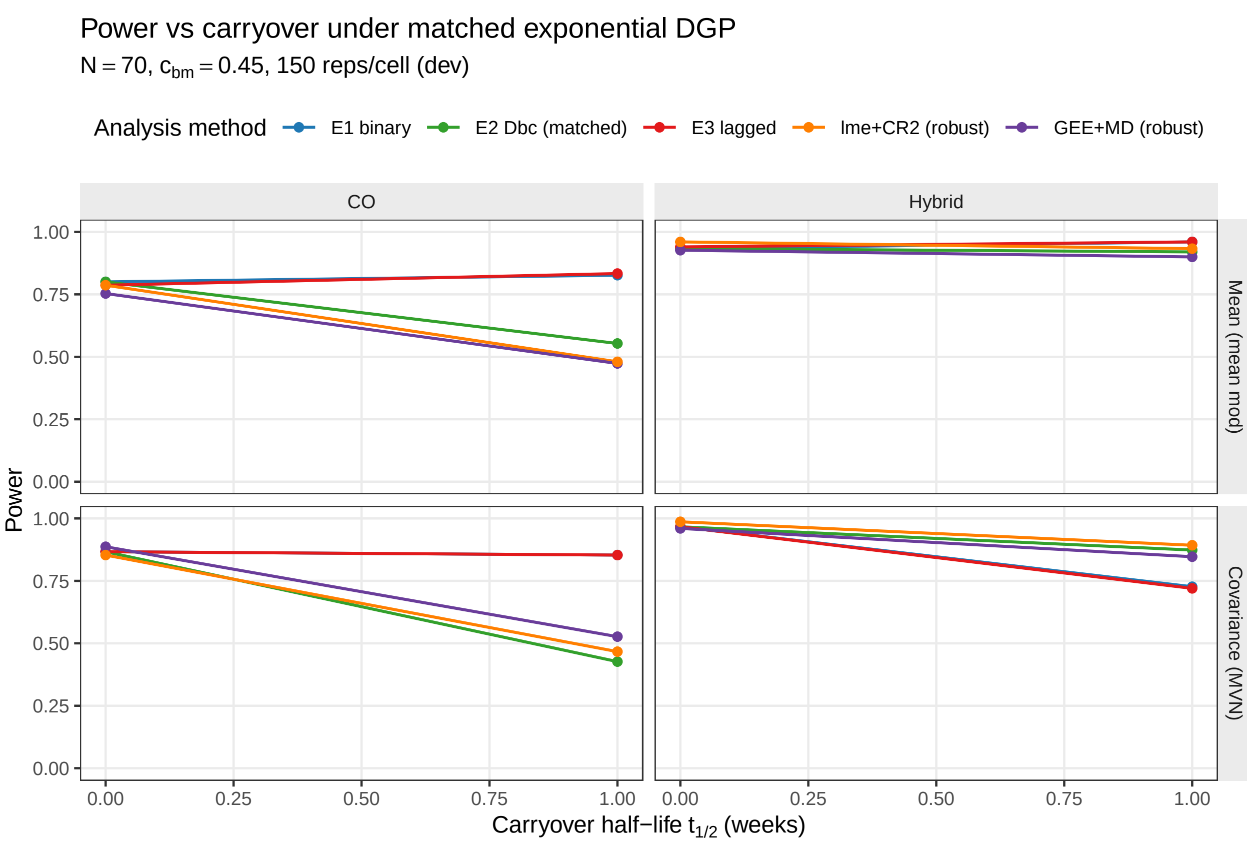

Figure 1: Power versus carryover half-life under the exponential DGP, stratified by architecture and design, for all five analysis methods. Dev grid: design in {CO, Hybrid} only; $N = 70$, $c_{bm} = 0.45$, 150 reps per cell. lme+CR2 lifts power above E2 by recalibrating the conservative standard error; GEE+MD tracks just below E2, a modest residual efficiency cost of the marginal estimator relative to the conditional lme.

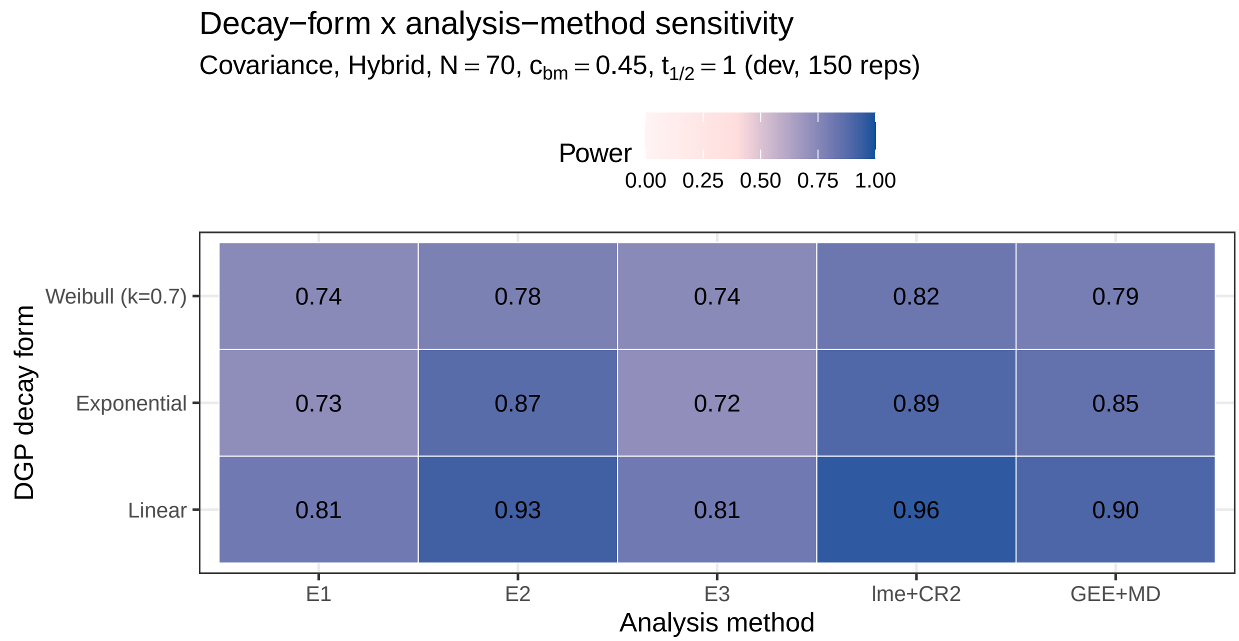

Figure 2: Power at the reference carryover cell ($t_{1/2} = 1.0$ week, the covariance-moderation architecture, Hybrid design, $N = 70$, $c_{bm} = 0.45$) across DGP decay forms and the five analysis methods. The dev grid contains three DGP forms: Linear, Exponential, and Weibull $k = 0.7$.

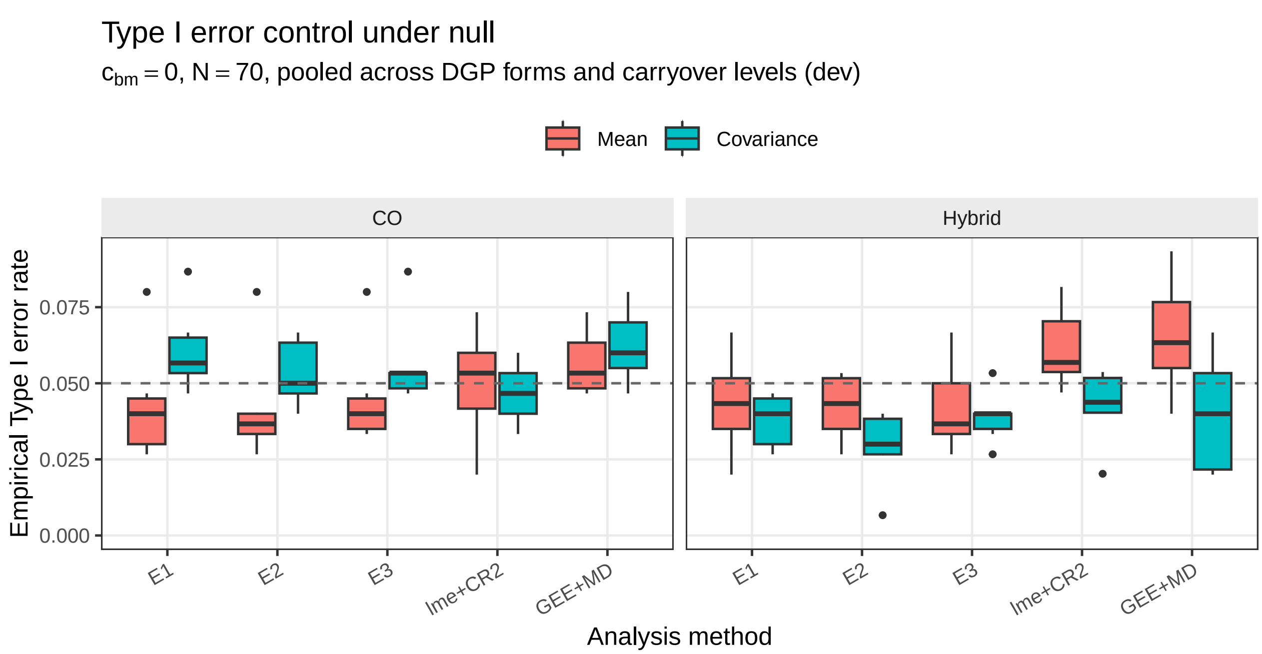

Figure 3: Empirical Type I error rates under the null ($c_{bm} = 0$), pooled across DGP decay forms and carryover half-lives, stratified by design and analysis method. $N = 70$; dashed line at nominal 0.05. The three model-based specifications sit below nominal; lme+CR2 sits near nominal; GEE+MD is near nominal under the Hybrid design but anti-conservative under CO.

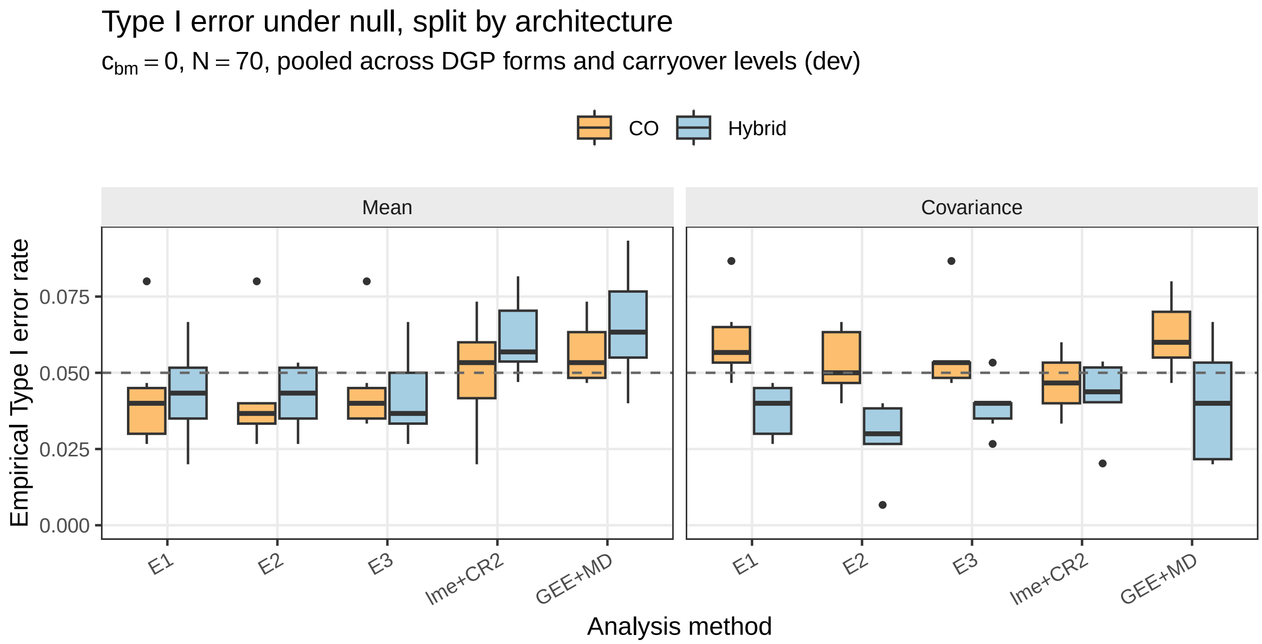

Figure 4: Empirical Type I error rates under the null ($c_{bm} = 0$, $N = 70$), the same null cells as the preceding figure but split by DGP architecture, with design as the fill. At the null the two architectures are a bit-identical data-generating process, so the two panels coincide up to Monte Carlo noise. Dashed line at nominal 0.05.

Table 3: Power and non-convergence at $N = 70$, $c_{bm} = 0.45$, Hybrid design, dev grid (150 reps per cell), for all five analysis methods across the DGP-architecture combinations present in the dev grid.

Arch	DGP	t1/2	spec	Power (MCSE)	Non-conv
A	exponential	0	E1	0.933 (0.020)	0.000
A	exponential	0	E2	0.933 (0.020)	0.000
A	exponential	0	E3	0.940 (0.019)	0.000
A	exponential	0	lme+CR2	0.960 (0.016)	0.000
A	exponential	0	GEE+MD	0.927 (0.021)	0.000
A	exponential	1	E1	0.960 (0.016)	0.000
A	exponential	1	E2	0.920 (0.022)	0.000
A	exponential	1	E3	0.960 (0.016)	0.000
A	exponential	1	lme+CR2	0.933 (0.020)	0.007
A	exponential	1	GEE+MD	0.900 (0.024)	0.000
A	linear	0	E1	0.967 (0.015)	0.000
A	linear	0	E2	0.967 (0.015)	0.000
A	linear	0	E3	0.967 (0.015)	0.000
A	linear	0	lme+CR2	0.966 (0.015)	0.007
A	linear	0	GEE+MD	0.947 (0.018)	0.000
A	linear	1	E1	0.967 (0.015)	0.000
A	linear	1	E2	0.867 (0.028)	0.000
A	linear	1	E3	0.960 (0.016)	0.000
A	linear	1	lme+CR2	0.886 (0.026)	0.007
A	linear	1	GEE+MD	0.880 (0.027)	0.000

A	weibull k=0.7	0	E1	0.993 (0.007)	0.000
A	weibull k=0.7	0	E2	0.993 (0.007)	0.000
A	weibull k=0.7	0	E3	1.000 (0.000)	0.000
A	weibull k=0.7	0	lme+CR2	0.993 (0.007)	0.007
A	weibull k=0.7	0	GEE+MD	0.960 (0.016)	0.000
A	weibull k=0.7	1	E1	0.973 (0.013)	0.000
A	weibull k=0.7	1	E2	0.947 (0.018)	0.000
A	weibull k=0.7	1	E3	0.973 (0.013)	0.000
A	weibull k=0.7	1	lme+CR2	0.973 (0.013)	0.000
A	weibull k=0.7	1	GEE+MD	0.927 (0.021)	0.000
B	exponential	0	E1	0.967 (0.015)	0.000
B	exponential	0	E2	0.967 (0.015)	0.000
B	exponential	0	E3	0.967 (0.015)	0.000
B	exponential	0	lme+CR2	0.987 (0.009)	0.007
B	exponential	0	GEE+MD	0.960 (0.016)	0.000
B	exponential	1	E1	0.727 (0.036)	0.000
B	exponential	1	E2	0.873 (0.027)	0.000
B	exponential	1	E3	0.720 (0.037)	0.000
B	exponential	1	lme+CR2	0.893 (0.025)	0.007
B	exponential	1	GEE+MD	0.847 (0.029)	0.000
B	linear	0	E1	0.980 (0.011)	0.000
B	linear	0	E2	0.980 (0.011)	0.000
B	linear	0	E3	0.980 (0.011)	0.000
B	linear	0	lme+CR2	0.980 (0.012)	0.013
B	linear	0	GEE+MD	0.953 (0.017)	0.000
B	linear	1	E1	0.813 (0.032)	0.000
B	linear	1	E2	0.927 (0.021)	0.000
B	linear	1	E3	0.813 (0.032)	0.000
B	linear	1	lme+CR2	0.959 (0.016)	0.013
B	linear	1	GEE+MD	0.900 (0.024)	0.000
B	weibull k=0.7	0	E1	1.000 (0.000)	0.000
B	weibull k=0.7	0	E2	1.000 (0.000)	0.000
B	weibull k=0.7	0	E3	1.000 (0.000)	0.000
B	weibull k=0.7	0	lme+CR2	1.000 (0.000)	0.020
B	weibull k=0.7	0	GEE+MD	0.987 (0.009)	0.000
B	weibull k=0.7	1	E1	0.740 (0.036)	0.000
B	weibull k=0.7	1	E2	0.780 (0.034)	0.000
B	weibull k=0.7	1	E3	0.740 (0.036)	0.000
B	weibull k=0.7	1	lme+CR2	0.824 (0.031)	0.013
B	weibull k=0.7	1	GEE+MD	0.793 (0.033)	0.000

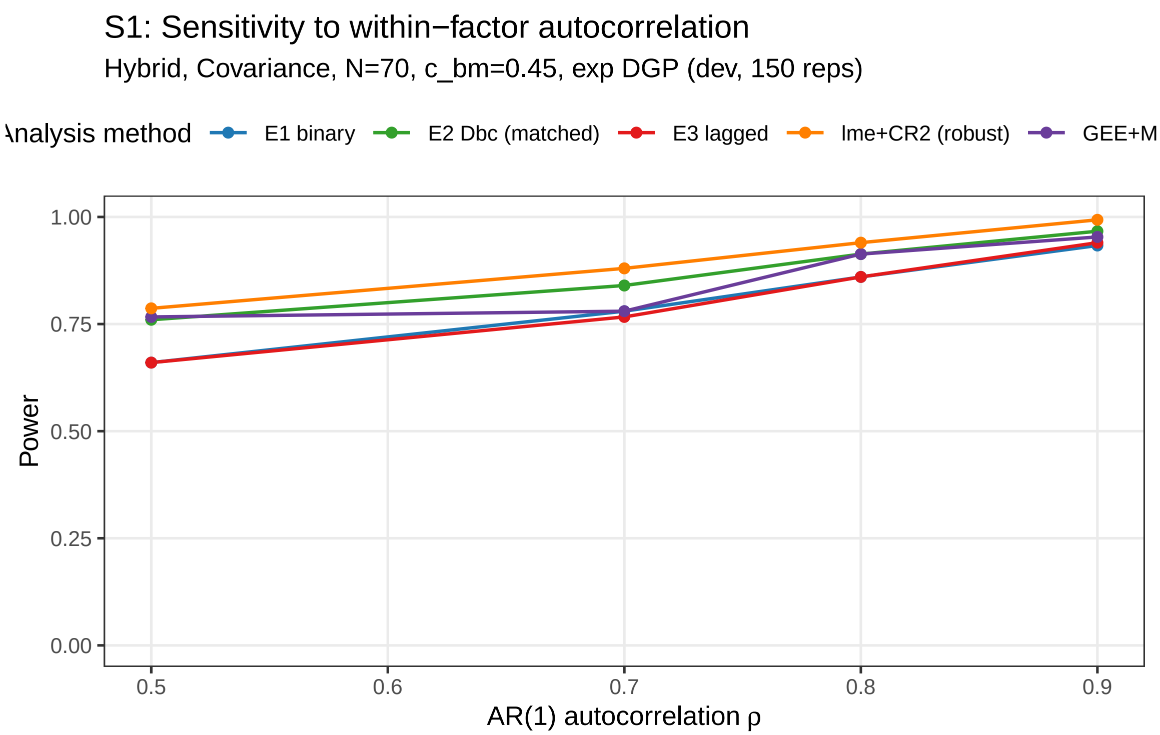

Figure 5: Power by analysis method across AR(1) autocorrelation $\rho \in \{0.5, 0.7, 0.8, 0.9\}$. Reference: Covariance, Hybrid, $N = 70$, $c_{bm} = 0.45$, exponential DGP, $t_{1/2} = 1.0$ weeks.

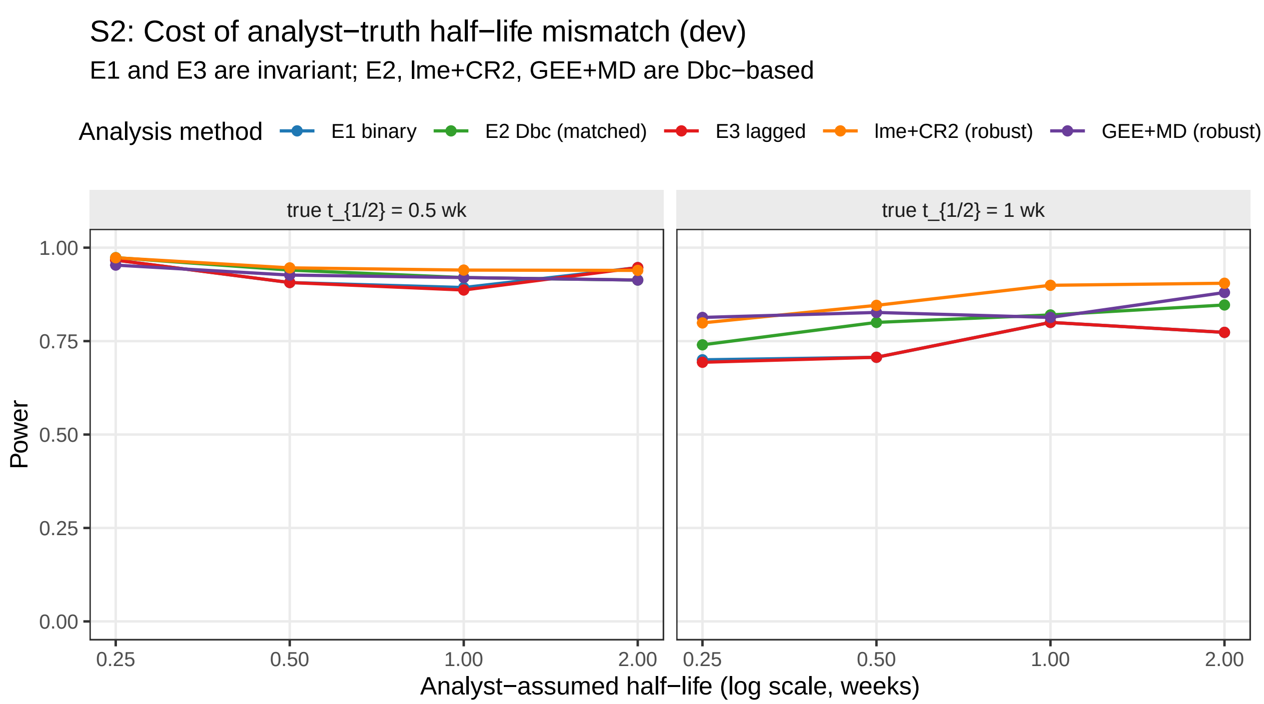

Figure 6: Power as the analyst's assumed half-life departs from the true DGP half-life. E1 and E3 do not consume the assumed half-life and are flat; E2 and the two robust analyses are all built on the D_{bc} predictor and vary with it.

Table 4: Automated sanity checks against the dev-run results. Thresholds are deliberately loose (MCSE 0.041 at 150 reps).

Check	Result
E2 power > E1 power at the reference cell	E2=0.873, E1=0.727 -> PASS
lme+CR2 power >= E2 power (robust SE recovers power)	lme+CR2=0.893, E2=0.873 -> PASS
lme+CR2 null Type I within 0.015 of nominal 0.05	lme+CR2 mean Type I=0.050 -> PASS
Max non-convergence rate below 0.20	Max=0.020 -> PASS

3 Tier 2 sensitivity results

3.1 S1: within-factor autocorrelation

3.2 S2: analyst-truth half-life mismatch

3.3 S3: dropout

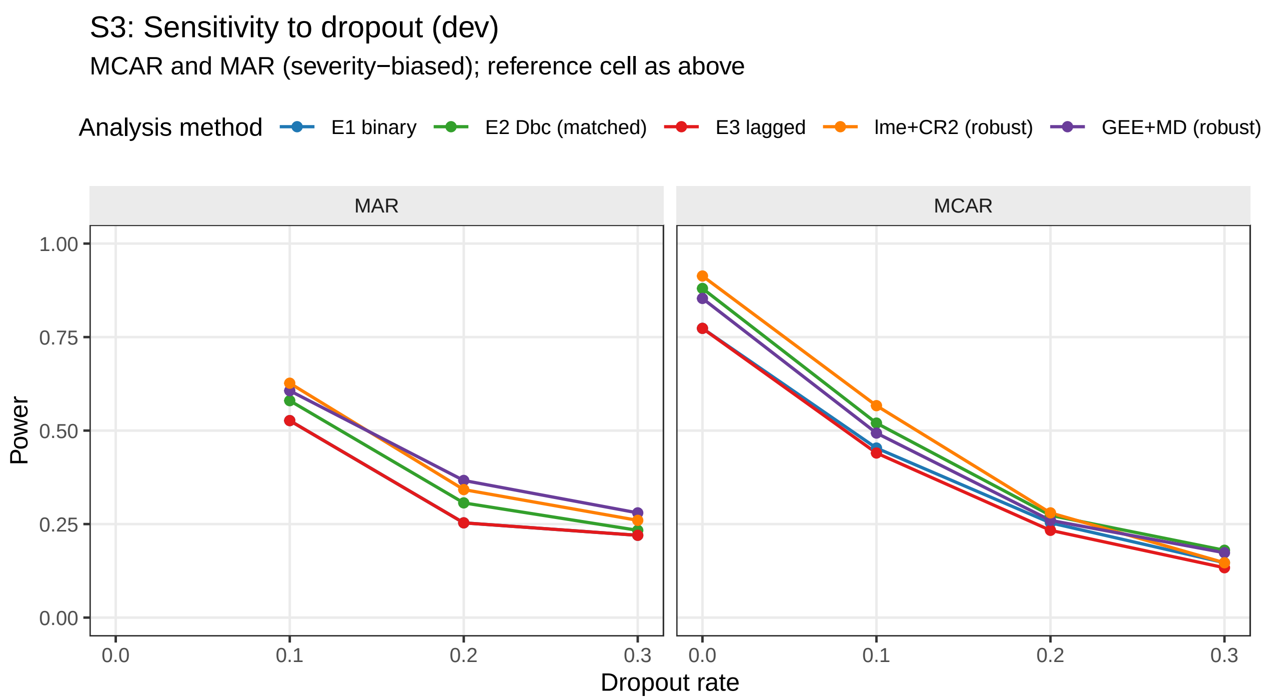

Figure 7: Power at dropout rates $\in \{0, 0.1, 0.2, 0.3\}$ under MCAR and MAR mechanisms, for all five analysis methods. With its exchangeable working correlation GEE+MD runs cleanly across the dropout range, power declining monotonically as expected.

3.4 S4: biomarker-moderation effect-size curve

3.5 S5: autocorrelation by carryover (exploratory)

4 Dev-run review checklist

Dev review document rendered from `analysis/report/02-carryover-sensitivity/report-devresults.Rmd` *on 2026-07-31 at 11:21 PDT. Source data:* `output/01-dev.rds` *(72000 rows),* `output/04-sensitivity-dev.rds` *(23250 rows).*

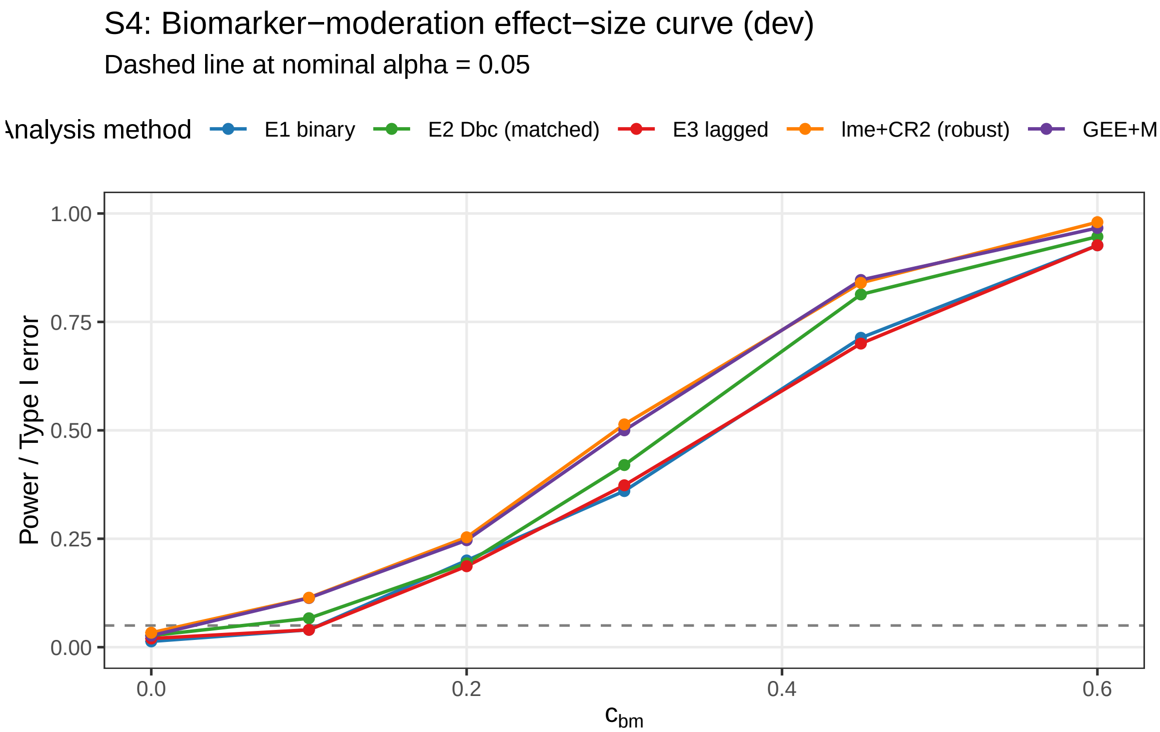

Figure 8: Power across $c_{bm} \in \{0, 0.10, 0.20, 0.30, 0.45, 0.60\}$, for all five analysis methods. The $c_{bm} = 0$ point is the null; dashed line at nominal $\alpha = 0.05$.

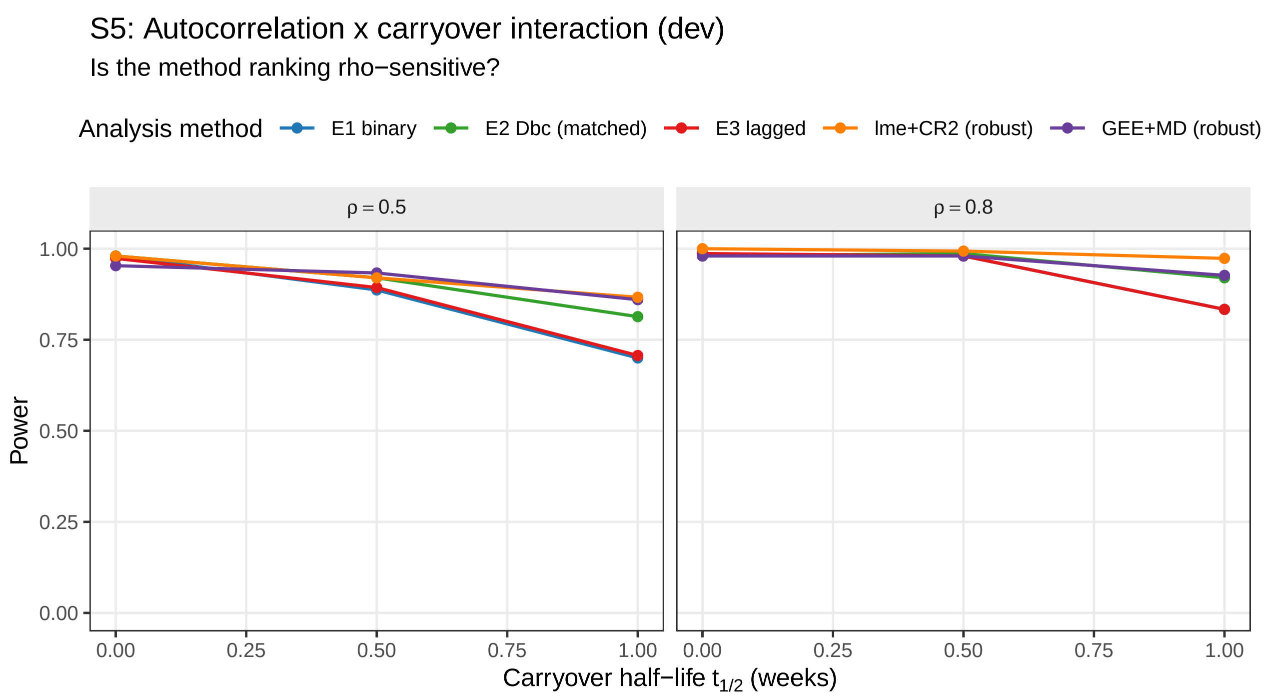

Figure 9: Power across carryover half-lives $\{0, 0.5, 1.0\}$ weeks at $\rho \in \{0.5, 0.8\}$, for all five analysis methods. Facets by ρ.